Quiz

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Midterm Exam due Jan 27, 2025 03:20 IST

Main notions learned in this chapter

5/6 points (graded)

The complexity difference $C(n+1) - C(n)$ can become arbitrarily large when the integer n grows.

- ○ True
- ◉ False

✔

Answer Correct:
We may find a generic way to add a final instruction such as "add 1" to any program that generates n. So the minimal program that generates $n+1$ cannot be much longer than the minimal program that generates \n\).

The complexity of x is defined as the average length over all programs that generate x.

- ○ True
- ◉ False

✔

Answer Correct: Description complexity is defined as the size of the shortest program.

If we knew a program p such that $C_U(x_0) = |p|$ for a given object x_0, could we *compute* the value of $C_V(x_0)$ for another universal machine V different from U?

- ◉ Yes
- ○ No

✖

Answer Incorrect: No. We can only find upper bounds for $C_V(x_0)$.

Suppose that only two programs p_1 and p_2 can generate string x on the reference machine. p_1 requires n^2 milliseconds to run while p_2 requires $n \times log(n)$ milliseconds. Then $C(x) = |p_2|$.

- ○ True
- ◉ False

✔

Answer Correct:
False! Description complexity is not related to the execution time but to the length of programs.

For any finite object x, there exists a machine on which the complexity of x is equal to 1.

- ◉ True

True

False

Answer
Correct: Such a machine can be built by permuting the order of programs.

$C(x|x) = 0.$

True

False

Answer
Correct: The complexity of x is zero just after x has been input.

Submit You have used 1 of 1 attempt

✱ Partially correct (5/6 points)

Problem 1

0.0/2.0 points (graded)

Geometrical figures can be represented with numbers. For instance, two numbers are sufficient to represent a straight line in the plane: the value of x and of y where the line intersects the x-axis and the y-axis. We suppose that the plane is finite and that all numbers can be represented with 10 bits each.

How many bits are needed at most to represent a square in that finite plane?

30 ✖

30

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Problem 2

2.0/2.0 points (graded)

Imagine a social network in which each person has at most k friends. Your friends are at distance 1 from you, your friend's friends at distance 2, and so on. How does people's complexity (in bits) grow at most as a function of distance d?

$d \times \log_2(k)$

$k \times \log_2(d)$

$\log_2(k \times d)$

$\log_2(k) + \log_2(d)$

Answer

Answer
Correct:
In a wide population, the worst situation is that you have k^d different people at distance d from you. You need $d \times \log_2(k)$ bits to discriminate them.

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Problem 3

2/2 points (graded)

A weekly lottery issues a 6-digit number between 000000 and 999999. We suppose that all previous draws are stored in memory. Today's draw is 767433. It was 767439 three weeks ago. More generally, what is a good estimate of the complexity of today's draw if it differs by d from the kth previous draw? ($\log_{2+}$ designates the logarithm in base 2 rounded to the upper integer and $O(1)$ designates a constant)

- $\log_{2+}(10^6) + O(1)$
- $\log_{2+}(d) + \log_{2+}(k) + O(1)$ ✔
- (selected) $min(\log_{2+}(k \times d), \log_{2+}(10^6)) + O(1)$
- $max(\log_{2+}(k \times d), \log_{2+}(10^6)) + O(1)$

✔

Answer
Correct:
We need $\log_{2+}(k) + \log_{2+}(d) + O(1)$ bits to retrieve a previous draw at distance k in memory and to tell the difference (the constant $O(1)$ includes the sign (positive or negative) of the difference). This method for describing the draw should not be more complex than a direct description. So $min(\log_{2+}(k \times d), \log_{2+}(10^6)) + O(1)$ is the correct answer.

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

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